

U.S. Weather Risk Analysis

2019-2024 | Google BigQuery + SQL | NOAA GSOD

5.64M

station-day observations

50

U.S. states

2,768

weather stations

0

duplicate station-date rows

OBJECTIVE & APPROACH

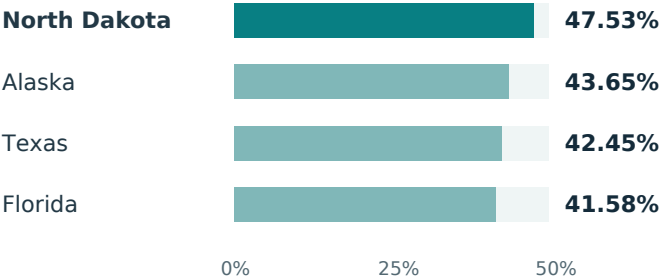
Identify state-level patterns in unusually warm, cold, wet and windy weather to guide deeper preparedness analysis. A reusable SQL view validates **5,638,214 observations** and ranks states by both frequency and count using national, dataset-derived percentile thresholds.

Data Quality Control: Resolved **NULLIF** type/sentinel mismatches with **SAFE_CAST**; recalculated **APPROX_QUANTILES** on positive rain days (>0.00 in), correcting the precipitation cutoff from **0.14 in to 0.78 in**.

KEY FINDINGS

Combined exposure: leading states

Share of station-days meeting at least one selected condition



North Dakota: 47.53% led proportionally; **Texas: 187,161** had the largest count among leading states.

LEADERS BY CONDITION

Condition / Cutoff	Leading State (% Share)
Warm: ≥79.2°F	Florida 35.15%
Cold: ≤29.4°F	North Dakota 32.65%
Rain: ≥0.78 in	Arkansas 5.51%
Wind: ≥13.23 mph	North Dakota 24.67%

Rain and wind shares use valid observations; temperature uses all observations.

Annual pattern: 2024 recorded the highest annual average temperature in the period (**56.46°F**), versus **54.51°F** in 2022 (**a 1.95°F spread across the six-year period**). Precipitation and wind averages varied within narrow ranges.

RECOMMENDED PRIORITIES

- 01 Target heat and winter screening.** Focus heat analysis on Florida and Texas; examine cold readiness in North Dakota and Alaska, adding population and infrastructure exposure.
- 02 Investigate rain and wind resilience.** Review Arkansas and Alabama for drainage needs, and North Dakota, South Dakota and Kansas for wind exposure using local hazard data.
- 03 Use combined exposure to guide deeper study.** Start with North Dakota, Alaska, Texas and Florida; assess each state’s hazard mix, vulnerability and recorded impacts.
- 04 Strengthen monitoring and quality controls.** Extend the time series; automate anomaly, duplicate and missingness checks, and assess changes in station coverage.

How to read the results Warm, rain and wind use the 90th percentile; cold uses the 10th. Temperature and wind cutoffs use daily averages. The rain cutoff is calculated from positive-rain observations. A station-day is one station on one date. Results are unweighted observation frequencies, not statewide days, official warnings or a severity score. Six years do not establish a long-term climate trend.